```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main(int argc, char *argv[])
{
 int mb;

 if (argc != 2){
printf("Usage:%s <memory_in_MB>\n", argv[0]);
return 1;
}

mb = atoi(argv[1]);
char *memory = malloc(mb * 1024 *1024);

if (memory == NULL)
{
printf("Memory allocation failed\n");
return 1;
}

for (int i=0; i < mb * 1024 * 1024; i += 4096){
memory[i] = 1;
}

printf("%d MB memory allocated.\n", mb);
printf("Program is running... Press Ctrl+C to exit.\n");

while (1){
sleep(1);
}


return 0;
}
```

```
vboxuser@Parul: ~ — watch -n 1 free -h

Every 1.0s: free -h                                          Parul: Mon Oct 12 16:22:10 2026

               total        used        free      shared  buff/cache   available
Mem:           2.1Gi       1.3Gi       284Mi        51Mi       832Mi       907Mi
Swap:             0B          0B          0B
```

```
vboxuser@Parul:~$ top

top - 16:23:00 up  2:19,  1 user,  load average: 0.61, 0.27, 0.19
Tasks: 261 total,   1 running, 260 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.2 us,  0.2 sy,  0.0 ni, 99.6 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem :   2200.2 total,    284.3 free,   1292.7 used,    832.7 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    907.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   3390 vboxuser  20   0 6213608 364836 118492 S   3.9  16.2   5:57.44 gnome-shell
   4802 vboxuser  20   0 3708456 331568  97260 S   1.3  14.7   3:46.78 ptyxis
   4822 vboxuser  20   0  239944   8012   7292 S   0.3   0.4   0:04.64 ptyxis-agent
   4986 root      20   0       0      0      0 I   0.3   0.0   0:00.37 kworker/u28:0-events_unbound
   5377 root      20   0       0      0      0 I   0.3   0.0   0:01.87 kworker/u26:2-events_freezable_pwr_efficient
   7191 root      20   0       0      0      0 I   0.3   0.0   0:00.33 kworker/1:1-events
      1 root      20   0   26204  16988  11712 S   0.0   0.8   0:10.94 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.34 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
```

```
vboxuser@Parul:~$ nano exp6.c
vboxuser@Parul:~$ gcc exp6.c -o exp6
vboxuser@Parul:~$ free
               total        used        free      shared  buff/cache   available
Mem:         2252996     1325952      289240       52412      852364      927044
Swap:              0           0           0
vboxuser@Parul:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           2.1Gi       1.3Gi       282Mi        51Mi       832Mi       905Mi
Swap:             0B          0B          0B
vboxuser@Parul:~$ watch -n 1 free -h
```